

Adachi and Shimamura

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Varieties

Affine varieties

Definition 1.1 (Algebraic sets) Let k be an algebraically closed field. We say $X \subseteq \mathbb{A}_k^n$ is an *algebraic set* if $X = Z(T)$ for some $T \subseteq k[x_1, \dots, x_n]$. Notice that $Z(T) = Z(\langle T \rangle) = Z(\sqrt{\langle T \rangle})$; so if X is an algebraic set we may assume WLOG that $X = Z(\mathfrak{a})$ for some radical ideal $\mathfrak{a} \subseteq k[x_1, \dots, x_n]$.

Proposition 1.2 (Zariski topology on $\mathbb{A}^n$) *The finite union and arbitrary intersection of algebraic sets are again algebraic sets; we define the Zariski topology on $\mathbb{A}_k^n$ by declaring the closed sets to be precisely the algebraic sets. In particular,*

$$\bigcup_{i=1}^n Z(\mathfrak{a}_i) = Z(\mathfrak{a}_1 \cdots \mathfrak{a}_n) \quad \text{and} \quad \bigcap_{i \in I} Z(\mathfrak{a}_i) = Z\left(\sum_i \mathfrak{a}_i\right).$$

- Notice that $\mathbb{A}_k^n = Z(0)$ and $\emptyset = Z(1)$ are both closed and open.

Example 1.2.1 (Zariski topology on $\mathbb{A}^1$) Since $k[x]$ is a PID, every proper non-empty algebraic set is of the form $Z(\langle f \rangle)$ for some $f \in k[x]$. But k is algebraically closed, so $f(x) = \prod_{i=1}^n (x - p_i)$ and therefore $Z(\langle f \rangle) = \{p_1, \dots, p_n\}$. Conversely, every finite set $S \subseteq \mathbb{A}^1$ corresponds to a function in $k[x_1, \dots, x_n]$. Hence the closed sets on $\mathbb{A}^1$ consist of $\mathbb{A}^1$, $\emptyset$, and all finite sets. In particular, the Zariski topology on $\mathbb{A}^1$ is not Hausdorff.

Definition 1.3 (Irreducible subsets) Let X be a topological space. We say a subset $Y \subseteq X$ is *irreducible* if $Y \neq Y_1 \cup Y_2$ where $Y_i \subsetneq Y$ is a proper closed subset.

- Since we care about algebraic sets which are defined to be closed, we want to understand how a large algebraic set may be decomposed into smaller ones.

Lemma 1.3.1 (Topological properties of irreducible sets) *Let $Y \subseteq X$ be irreducible. Then*

(1). *Any open subset of Y is dense and irreducible.*

(2). *$\overline{Y} \subseteq X$ is irreducible.*

- Hence irreducibility is a nice property that “passes” up to closure and down to open subsets.

Definition 1.3.2 (Affine varieties) We say an algebraic set $X \subseteq \mathbb{A}_k^n$ is an *affine variety* if X is irreducible; an open subset $Y \subseteq X$ is called a *quasi-affine variety*.

- We will show that Y is not just an open subset but carries other interesting structures.

Proposition 1.4 (Properties of Zariski topology) *If $Y \subseteq \mathbb{A}^n$ is any subset, we define $I(Y) \trianglelefteq A := k[x_1, \dots, x_n]$ to be the ideal generated by functions vanishing on Y . Then*

- (1). *If $Y_1 \subseteq Y_2$, then $I(Y_2) \subseteq I(Y_1)$.*
- (2). *If $T_1 \subseteq T_2 \subseteq A$, then $Z(T_2) \subseteq Z(T_1)$.*
- (3). *$I(Y_1 \cup Y_2) = I(Y_1) \cap I(Y_2)$.*
- (4). *If $\mathfrak{a} \trianglelefteq A$, then $I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.*
- (5). *$Z(I(Y)) = \overline{Y}$.*

- (1) - (3) are immediate and (4) follows from Nullstellensatz. For (5), notice that $Z(I(Y)) \supseteq \overline{Y}$ since it is a closed set containing Y . Conversely, if $p \notin \overline{Y}$, then $p \notin Y'$ for some closed Y' containing Y and hence $p \notin Z(I(Y')) \supseteq Z(I(Y))$.
- In particular, (4) implies that $V(\mathfrak{a}) \neq \emptyset$ for any proper $\mathfrak{a} \triangleleft A$.

Corollary 1.4.1 (Algebraic sets and radical ideals) *There are bijections*

$$\text{Rad } A \longleftrightarrow \{\text{algebraic sets in } \mathbb{A}^n\} \quad \text{and} \quad \text{Spec } A \longleftrightarrow \{\text{affine varieties in } \mathbb{A}^n\}.$$

- The first bijection follows from Proposition 1.4(4)(5). For the second one, suppose $\mathfrak{p} \in \text{Spec } A$ and

$$Z(\mathfrak{p}) =: X = X_1 \cup X_2 = Z(\mathfrak{a}_1) \cup Z(\mathfrak{a}_2) = Z(\mathfrak{a}_1 \mathfrak{a}_2).$$

Then we must have $\mathfrak{p} = \sqrt{\mathfrak{p}} = \sqrt{\mathfrak{a}_1 \mathfrak{a}_2} \supseteq \sqrt{\mathfrak{a}_1} \sqrt{\mathfrak{a}_2}$ and so $\mathfrak{p} \supseteq \sqrt{\mathfrak{a}_1}$ or $\mathfrak{p} \supseteq \sqrt{\mathfrak{a}_2}$, which implies that $X = X_1$ or $X = X_2$ and it is irreducible. Conversely, suppose $X = Z(\mathfrak{q})$ is irreducible and $\mathfrak{q} \supseteq \mathfrak{a}\mathfrak{b}$. Then $X \subseteq Z(\mathfrak{a}\mathfrak{b}) = Z(\mathfrak{a}) \cup Z(\mathfrak{b})$. Now since $X = (X \cap Z(\mathfrak{a})) \cup (X \cap Z(\mathfrak{b}))$ is irreducible, we may assume WLOG that $X = X \cap Z(\mathfrak{a})$, that is, $\sqrt{\mathfrak{q}} = \sqrt{\mathfrak{q} + \mathfrak{a}}$; but that implies $\mathfrak{q} = \mathfrak{a}$ and therefore $\mathfrak{q}$ is prime.

Example 1.4.2 Since $A = k[x_1, \dots, x_n]$ is an integral domain, $\mathbb{A}_k^n$ is irreducible. Suppose $f \in A$ is an irreducible polynomial of degree d ; if $n = 2$ we say $Z(f)$ is an affine curve of degree d ; if $n \geq 3$ we say $Z(f)$ is a hypersurface.

Definition 1.5 (Affine coordinate ring) If Y is an affine variety, we define its *affine coordinate ring* to be $A(Y) := A/I(Y)$ which is an integral domain.

- Hence there is a bijection between finitely generated k -domains and affine varieties.

Definition 1.6 (Noetherian spaces) We say a topological space X is *Noetherian* if it satisfies the descending chain condition (DCC) on closed sets, or equivalently it satisfies the ascending chain condition (ACC) on open sets.

Example 1.6.1 $\mathbb{A}_k^n$ is a Noetherian space since every closed set corresponds to an ideal in $k[x_1, \dots, x_n]$, which is a Noetherian ring.

Proposition 1.6.2 (Properties of Noetherian spaces) *Let X be a Noetherian space. Then every non-empty closed subset $Y \subseteq X$ is a finite union of irreducible closed subsets. In particular, we can write $Y = \bigcup_{i=1}^n Y_i$ where $Y_i \not\subseteq Y_j$ if $i \neq j$; such a decomposition is unique and the Y_i 's are called the irreducible components of Y .*

- This is completely analogous to that fact that every non-empty ideal in a Noetherian ring is a finite intersection of irreducible ideals, and the proof is also similar.

Corollary 1.6.3 (Algebraic sets and affine varieties) *Every algebraic set is a finite union of affine varieties, none containing another.*

- In fact, we can be more precise: if $Z(\mathfrak{a}) = \bigcup_{i=1}^n Z(\mathfrak{p}_i)$ where each $Z(\mathfrak{p}_i)$ is an affine variety, then the $\mathfrak{p}_i$'s are *precisely* the minimal prime ideals over $\mathfrak{a}$. See Exercise I.1.9.

Definition 1.7 (Dimension of topological spaces) We define the dimension of a topological space X to be

$$\dim X := \sup\{n \in \mathbb{N} : \exists Z_0 \subsetneq \cdots \subsetneq Z_n \subseteq X \text{ where } Z_i \text{ is irreducible closed}\}.$$

Proposition 1.7.1 (Dimension of algebraic sets) *Let $Y \subseteq \mathbb{A}_k^n$ be an algebraic set. Then $\dim Y = \dim A(Y)$. In particular, $\dim \mathbb{A}_k^n = \dim k[x_1, \dots, x_n] = n$.*

- This is a consequence of Corollary 1.4.1 and the correspondence theorem.

Proposition 1.7.2 (Dimension of quasi-affine varieties) *Let Y be a quasi-affine variety. Then $\dim Y = \dim \bar{Y}$.*

- This follows from a few topological facts. If $Z_1 \subsetneq Z_2$ are closed subsets in Y , then $\bar{Z}_1 \subsetneq \bar{Z}_2$ in $\bar{Y}$, since $Z_i = \bar{Z}_i \cap Y$; so then $\dim Y \leq \dim \bar{Y}$. Now by Proposition 1.7.1 and the correspondence theorem, $\dim \bar{Y} = \dim A(\bar{Y}) \leq \dim \mathbb{A}^n = n$, so we can pick a maximal chain $Z_0 \subsetneq \cdots \subsetneq Z_m$ in Y . We claim that this corresponds to a maximal chain $\bar{Z}_0 \subsetneq \cdots \subsetneq \bar{Z}_m$ in $\bar{Y}$. Indeed, notice that $\bar{Z}_0$ is a point since $I(\bar{Z}_0)$ corresponds to a maximal ideal in $A(\bar{Y})$; moreover, since Y is irreducible in $\bar{Y}$ by Lemma 1.3.1, we deduce $Z_m = Y$ and therefore $\bar{Z}_m = \bar{Y}$. Finally, since $A(\bar{Y})$ is finitely generated k -domain, its dimension is the same as the height of any maximal ideal in it; hence $\dim Y = m = n = \dim \bar{Y}$.

Proposition 1.7.3 (Varieties of codimension one) *A variety $Y \subseteq \mathbb{A}^n$ is of codimension 1 $\iff Y = Z(f)$ for some irreducible polynomial $f \in A$.*

- This relies on the following result:

A Noetherian domain is a UFD $\iff$ every prime ideal of height 1 is principal.

Recall that $\dim Y = \text{ht}(\mathfrak{m})$ for every $\mathfrak{m} \in \text{Max } A(Y)$. Since $A = k[x_1, \dots, x_n]$ is a UFD, every prime ideal of height 1 is principal, so they are all of the form $\langle f \rangle$ for some irreducible polynomial $f \in A$. Now by the correspondence theorem, every maximal chain of prime ideals in $A(Y)$ corresponds to a sub-chain of a maximal chain in A . Hence

$$\dim A(Y) = \dim A - \text{ht}(I(Y)) = n - 1 \iff \text{ht}(I(Y)) = 1 \implies I(Y) = \langle f \rangle$$

for some irreducible $f \in A$. Conversely, if $\mathfrak{p} = \langle f \rangle \in \text{Spec } A$, then $\text{ht}(\mathfrak{p}) = 1$ by Krull's PIT.

Exercise I.1

- 1.1** (a) Let $Y = Z(x^2 - y) \subseteq \mathbb{A}_k^2$. Show that $A(Y) \simeq k[t]$.
 (b) Let $Y' = Z(xy - 1) \subseteq \mathbb{A}_k^2$. Show that $A(Y') \not\simeq k[t]$.
 (c) Suppose $f \in k[x, y]$ is an irreducible quadratic polynomial and let $W = Z(f)$. Show that either $A(W) \simeq A(Y)$ or $A(W) \simeq A(Y')$.

Proof. (a) Consider $\varphi : k[x, y] \rightarrow k[t]$ via $f(x, y) \mapsto f(t, t^2)$. Clearly φ is a surjective k -algebra homomorphism. Notice that $\langle x^2 - y \rangle \subseteq \ker \varphi$. Conversely, if $f \notin \langle x^2 - y \rangle$, then by the division algorithm $f(x, y) = (x^2 - y)g(x, y) + r(x)$ where $r(x) \neq 0$. Hence $f(t, t^2) = r(t) \neq 0$ and $f \notin \ker \varphi$. So $A(Y) = k[x, y]/\langle x^2 - y \rangle \simeq k[t]$.

(b) Consider $\psi : k[x, y] \rightarrow k[t, t^{-1}]$ via $f(x, y) \mapsto f(t, t^{-1})$ which is clearly surjective. As in (a) we can show that $\ker \psi = \langle xy - 1 \rangle$. Hence $A(Y') = k[x, y]/\langle xy - 1 \rangle \simeq k[t, t^{-1}] \not\simeq k[t]$.

(c) Write $f(x, y) = f_2(x, y) + f_1(x, y) + f_0$ where f_i is of homogeneous degree i . Since k is algebraically closed, f_2 factors as $u(x, y)v(x, y)$ where u and v are of degree 1. Now if u and v are linearly independent, we may write f_1 as their linear combination. So then

$$f = uv + c_1u + c_2v + f_0 = (u + c_2)(v + c_1) + (f_0 - c_1c_2) =: u'v' + c'$$

where $c' \neq 0$ since f is irreducible. Hence by change of coordinate we deduce

$$A(W) = k[x, y]/\langle f \rangle \simeq k[u', v']/\langle u'v' - c' \rangle \simeq A(Y').$$

If u and v are linearly dependent, choose $v' \in k[x, y]$ to write f_1 as the linear combination of u and v' . By the end of day we deduce $f = du^2 - v''$ for some $d \in k$ and v'' in terms of u and v' . Hence

$$A(W) \simeq k[u, v'']/\langle du^2 - v'' \rangle \simeq A(Y).$$

□

- The factoring technique we use here only works for k adjoined with *at most two* variables. If $f = f_3 + f_2 + f_1 + f_0 \in k[x, y, z]$, then we cannot guarantee that f_3 and f_2 factor into linear terms.

- 1.2 Twisted Cubic Curve.** Let $Y = \{(t, t^2, t^3) : t \in k\} \subseteq \mathbb{A}_k^3$. Show that $\dim Y = 1$ and $A(Y) \simeq k[x]$.

Proof. Notice that

$$Y = Z(\langle y - x^2 \rangle) \cap Z(\langle z - x^3 \rangle) = Z(\langle y - x^2, z - x^3 \rangle).$$

It is clear that $\varphi : k[x, y, z] \rightarrow k[x]$ via $f(x, y, z) \mapsto f(x, x^2, x^3)$ is a surjective k -algebra homomorphism. We claim that $\ker \varphi = \langle y - x^2, z - x^3 \rangle =: \mathfrak{a}$. If $f \notin \mathfrak{a}$, then by the division algorithm,

$$f(x, y, z) = (y - x^2)g_1(x, y, z) + (z - x^3)g_2(x, y, z) + g_3(x)$$

where $g_3(x) \neq 0$. Hence $f(x, x^2, x^3) = g_3(x) \neq 0$ and so $f \notin \ker \varphi$. As a result, $A(Y) = k[x, y, z]/\mathfrak{a} \simeq k[x]$ is a PID, so $\dim Y = \dim A(Y) = 1$. □

- 1.3** Let $Y = Z(\langle x^2 - yz, xz - x \rangle) \subseteq \mathbb{A}^3$. Show that Y is the union of three irreducible components and describe their prime ideals.

Proof. We have

$$\begin{aligned} Y &= Z(\langle x^2 - yz \rangle) \cap Z(\langle xz - x \rangle) = Z(\langle x^2 - yz \rangle) \cap (Z(\langle x \rangle) \cup Z(\langle z - 1 \rangle)) \\ &= Z(\langle x^2 - yz, x \rangle) \cup Z(\langle x^2 - yz, z - 1 \rangle) = Z(\langle yz, x \rangle) \cup Z(\langle x^2 - y, z - 1 \rangle) \\ &= Z(\langle x, y \rangle) \cup Z(\langle x, z \rangle) \cup Z(\langle x^2 - y, z - 1 \rangle) \end{aligned}$$

where $\langle x, y \rangle$, $\langle x, z \rangle$, and $\langle x^2 - y, z - 1 \rangle$ are all prime ideals in $k[x, y, z]$. □

1.4 Show that the Zariski topology on $\mathbb{A}^2$ is not the product of two copies of Zariski topology on $\mathbb{A}^1$.

Proof. Recall that the only non-empty proper closed sets in $\mathbb{A}^1$ are finite sets. Hence any non-empty proper closed set in $\mathbb{A}^1 \times \mathbb{A}^1$ is a finite grid. But $Z(\langle x^2 - y \rangle) \subseteq \mathbb{A}^2$ is an infinite proper closed set, so these two topologies are not equal. $\square$

1.5 Show that a k -algebra B is the coordinate ring for some algebraic set in $\mathbb{A}_k^n \iff B$ is a finitely generated k -algebra with no nilpotent elements.

Proof. Recall that there is a bijection

$$\text{Rad } k[x_1, \dots, x_n] \leftrightarrow \{\text{algebraic sets in } \mathbb{A}_k^n\}.$$

Also for any ring A , $\mathfrak{a} \subseteq \text{Rad } A \iff A/\mathfrak{a}$ has no nilpotent elements, so we are done. $\square$

1.6 Prove Lemma 1.3.1.

1.7 Let X be a Noetherian space. Show that

- (a) X is quasi-compact; i.e., every open cover of X admits a finite subcover.
- (b) Any subset Y of X is Noetherian in the induced topology.
- (c) If X is Hausdorff then X must be a finite set with discrete topology.

Proof. (a) Suppose $\{U_i\}_{i \in I}$ is an open cover of X . Define $\Sigma := \{\bigcup_{j \in J} U_j : |J| < \infty\}$. Then Σ contains a maximal element $\bigcup_j U_j$ since X is Noetherian. We claim that $\bigcup_j U_j = X$; if not, since $X = \bigcup_i U_i$, we can form $U_i \cup \bigcup_j U_j \supsetneq \bigcup_j U_j$ for some U_i , contradicting maximality. Hence $\{U_j\}$ is a finite cover.

(b) This is clear since every ascending chain of open subsets in Y corresponds to an ascending chain of open sets in X .

(c) Since X is Noetherian, every subset of X is quasi-compact. Recall that a quasi-compact set in a Hausdorff space is closed, so every subset of X is closed and therefore open. Hence the topology on X is discrete; as a result, X must be finite because of the ascending chain condition. $\square$

1.8 Let $Y \subseteq \mathbb{A}^n$ be an affine variety of dimension r and $H \subseteq \mathbb{A}^n$ be a hypersurface. Suppose $Y \not\subseteq H$. Then every irreducible component of $Y \cap H$ has dimension $r - 1$.

Proof. Since H is a hypersurface, $H = Z(f)$ for some irreducible $f \in A := k[x_1, \dots, x_n]$ by Proposition 1.7.3. Notice that we can write

$$Y \cap H = Z(\mathfrak{p}) \cap Z(f) = Z(\mathfrak{p} + \langle f \rangle) = \bigcup_{i=1}^m Z(\mathfrak{p}_i)$$

for some $\mathfrak{p}, \mathfrak{p}_i \in \text{Spec } A$ where $\text{ht}(\mathfrak{p}) = n - \dim Y = n - r$. We claim that it suffices to show that $\text{ht}(\mathfrak{p}_i) = \text{ht}(\mathfrak{p}) + 1$, since then

$$\dim Z(\mathfrak{p}_i) = \dim A/\mathfrak{p}_i = n - \text{ht}(\mathfrak{p}_i) = n - (\text{ht}(\mathfrak{p}) + 1) = n - (n - r) - 1 = r - 1.$$

To this end, we show that each $\bar{\mathfrak{p}}_i$ is minimal over $\langle \bar{f} \rangle$ in the quotient ring $A/\mathfrak{p}$; notice that $\bar{f} \notin D(A/\mathfrak{p}) \cup (A/\mathfrak{p})^\times$ since $A/\mathfrak{p}$ is an integral domain and $Y \not\subseteq H$, meaning that $f \notin \mathfrak{p}$. But by the proof of Exercise 1.1.9 (see below), we know that the $\mathfrak{p}_i$'s are minimal over $\mathfrak{p} + \langle f \rangle$, so passing to $A/\mathfrak{p}$ the $\bar{\mathfrak{p}}_i$'s become minimal over $\langle \bar{f} \rangle$. Hence by Krull's PIT, $\text{ht}(\bar{\mathfrak{p}}_i) \leq 1$; but $\bar{0}$ is the unique minimal prime ideal and $\bar{\mathfrak{p}}_i \neq 0$, so $\text{ht}(\bar{\mathfrak{p}}_i) = 1$ and we are done. $\square$

1.9 Let $\mathfrak{a} \subseteq A := k[x_1, \dots, x_n]$ be an ideal which can be generated by r elements. Then every irreducible component of $Z(\mathfrak{a})$ has dimension $\geq n - r$.

Proof. Suppose $Z(\mathfrak{a}) = \bigcup_{i=1}^m Z(\mathfrak{p}_i)$. Then for every i ,

$$\dim Z(\mathfrak{p}_i) = \dim A/\mathfrak{p}_i \geq n - r \iff \text{ht}(\mathfrak{p}_i) \leq r.$$

Hence by Krull's HT, it suffices to show that each $\mathfrak{p}_i$ is minimal over $\mathfrak{a}$ which can be generated by r elements. But notice that

$$\sqrt{\mathfrak{p}_1 \cdots \mathfrak{p}_m} = \sqrt{\mathfrak{a}} = \bigcap_{\substack{\mathfrak{p}' \in V(\mathfrak{a}), \\ \mathfrak{p}': \text{minimal}}} \mathfrak{p}',$$

so every minimal $\mathfrak{p}'$ contains some $\mathfrak{p}_i$ and conversely every $\mathfrak{p}_i$ contains some minimal $\mathfrak{p}'$. Since the $\mathfrak{p}_i$'s do not contain each other (Corollary 1.6.3), we deduce the $\mathfrak{p}_i$'s are precisely the minimal prime ideals over $\mathfrak{a}$, so we are done. $\square$

1.10 Let X be a topological space.

- (a) If $Y \subseteq X$ is a subset, then $\dim Y \leq \dim X$.
- (b) If $\{U_i\}$ is an open cover of X , then $\dim X = \sup \dim_i U_i$.
- (c) Give an example of X with dense subset $U \subseteq X$ such that $\dim U < \dim X$.
- (d) If X is irreducible and finite-dimensional and $Y \subseteq X$ is a closed subset such that $\dim Y = \dim X$, then $Y = X$.
- (e) Give an example of an infinite-dimensional Noetherian space.

Proof. (a) This is clear by the proof of Proposition 1.7.2.

(b) It suffices to show that $\dim X \leq \sup \dim_i U_i$. Let $Z_0 \subsetneq \cdots \subsetneq Z_n$ be a chain of closed irreducible sets in X . Since $Z_0 \subseteq X$, we can pick $U \in \{U_i\}$ such that $Z_0 \cap U \neq \emptyset$. Now for every $0 \leq j \leq n$, define $Z'_j := U \cap Z_j$. We claim that $Z'_0 \subsetneq \cdots \subsetneq Z'_n$ is a chain of irreducible closed sets in U . Indeed, if $Z'_j \subseteq U$ is not irreducible, then $Z'_j = V_1 \cup V_2$ for some proper closed V_k ; but then $Z'_j = (V_1 \cap Z_j) \cup (V_2 \cap Z_j)$ is not irreducible in Z_j either, which contradicts Lemma 1.3.1 since $Z'_j \subseteq Z_j$ is an open subset. Finally, to see that $Z'_j \subsetneq Z'_{j+1}$, notice that each $Z'_j \subseteq Z_j$ is dense; so if $Z'_j = Z'_{j+1}$ then we must have $Z_j = \overline{Z'_j} = \overline{Z'_{j+1}} = Z_{j+1}$, a contradiction, so we are done.

(c) Consider a two-point space $X = \{p, q\}$ where the open sets are $\emptyset, \{p\}$, and X . Then $\{p\} \subseteq X$ is dense and $\dim\{p\} = 0$. But $\dim X = 1$ since $\{q\} \subsetneq X$ is irreducible and closed.

(d) Since $\dim Y = \dim X = n$, we can find a chain of irreducible closed subsets $Z_0 \subsetneq \cdots \subsetneq Z_n$ in Y . If $Y \neq X$, then $X \supsetneq Z_n$. But X itself is irreducible, so $Z_0 \subsetneq \cdots \subsetneq Z_n \subsetneq X$ is a chain of irreducible closed sets in X ; hence $\dim X \geq n + 1$, a contradiction.

(e) Let A be any infinite-dimensional Noetherian ring. We define a topology on $X = \text{Spec } A$ by declaring the closed sets to be precisely $\{V(\mathfrak{a})\}_{\mathfrak{a} \subseteq A}$. Then X is Noetherian since any descending chain $V(\mathfrak{a}_1) \supseteq V(\mathfrak{a}_2) \supseteq \cdots$ will correspond to $\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \cdots$ in A ; but A is Noetherian, so $\mathfrak{a}_n = \mathfrak{a}_{n+1} = \cdots$ for some $n \in \mathbb{N}$. Hence $V(\mathfrak{a}_n) = V(\mathfrak{a}_{n+1}) = \cdots$ and X satisfies the DCC on closed sets. $\square$

1.11 Let $Y = \{(t^3, t^4, t^5) : t \in k\} \subseteq \mathbb{A}^3$. Show that $I(Y)$ is a prime ideal of height 2 in $k[x, y, z]$ which cannot be generated by 2 elements. We say that Y is *not a local complete intersection* (See Exercise I.2.17).

1.12 Give an example of irreducible polynomial $f \in \mathbb{R}[x, y]$ such that $Z(f)$ is not irreducible in $\mathbb{A}_{\mathbb{R}}^2$.

Proof. $Z(x^2 + 1) = \emptyset \subseteq \mathbb{A}_{\mathbb{R}}^2$ is not irreducible. $\square$

Projective varieties

Definition 2.1 (Projective n -space) Let k be an algebraically closed field. We define an equivalent relation on $\mathbb{A}_k^{n+1} \setminus \{0\}$ by

$$(x_0, \dots, x_n) \sim (x'_0, \dots, x'_n) \iff (\exists \lambda \in k^\times) : (\lambda x_0, \dots, \lambda x_n) = (x'_0, \dots, x'_n)$$

and we say $\mathbb{P}^n := (\mathbb{A}_k^{n+1} \setminus \{0\}) / \sim$ is the *projective n -space over k* .

Definition 2.2 (Graded ring and homogeneous ideal) We say a ring S is *graded* if $S = \bigoplus_{d \in \mathbb{N}} S_d$ where each S_d is an abelian group and $S_d \cdot S_e \subseteq S_{d+e}$. The elements in each S_d are called *homogeneous elements* of S ; in particular, every $s \in S$ is a finite sum of homogeneous elements.

We say $\mathfrak{a} \subseteq S$ is a *homogeneous ideal* if $\mathfrak{a} = \bigoplus_{d \in \mathbb{N}} (\mathfrak{a} \cap S_d)$, or equivalently, $\mathfrak{a}$ is generated by homogeneous elements. Notice that the sum, product, intersection, and radical of homogeneous ideals are again a homogeneous ideal. To show that a homogeneous ideal $\mathfrak{a}$ is prime, it suffices to show that for all homogeneous elements $f, g \in S$ such that $fg \in \mathfrak{a}$, $f \in \mathfrak{a}$ or $g \in \mathfrak{a}$. Finally, if $\mathfrak{a} \subseteq S$ is a homogeneous ideal, then

$$S/\mathfrak{a} = \frac{\bigoplus_{d \in \mathbb{N}} S_d}{\bigoplus_{d \in \mathbb{N}} \mathfrak{a} \cap S_d} \simeq \bigoplus_{d \in \mathbb{N}} S_d / (\mathfrak{a} \cap S_d)$$

is again a graded ring.

Construction 2.3 (Polynomials over $\mathbb{P}^n$) Notice that $S = k[x_0, \dots, x_n]$ is a graded ring with each S_d defined to be the set of all monomials of total weight d in $x_0, \dots, x_n$. Pick a homogeneous element $f \in S$; so then $f(\lambda x_0, \dots, \lambda x_n) = \lambda^{\deg f} f(x_0, \dots, x_n)$.

We realize f as a function from $\mathbb{P}^n$ to $\{0, 1\}$ as follows: let $(x_0, \dots, x_n) \in \mathbb{A}_k^{n+1} \setminus \{0\}$ be a representative of a coset $p \in \mathbb{P}^n$, we define $f(p) = 0$ if $f(x_0, \dots, x_n) = 0$. This is well-defined, since for any other representative $p' = \lambda p$, if $f(p) = 0$ then $f(p') = f(\lambda p) = \lambda^{\deg f} f(p) = 0$; similarly if $f(p) \neq 0$, then $f(p') = \lambda^{\deg f} f(p) \neq 0$, in which case we simply define $f(p) = 1$.

We extend this construction to any $f \in S$ by defining $f(p) = 0 \iff f_i(p) = 0$ for every homogeneous component f_i of f . For a subset $T \subseteq S$, we define its zero locus to be $Z(T) = \{p \in \mathbb{P}^n : \forall f \in T, f(p) = 0\}$; and for a subset $Y \subseteq \mathbb{P}^n$, we define $I(Y)$ to be the ideal generated by $\{f \in S : \forall p \in Y, f(p) = 0\}$.

Definition 2.4 (Algebraic sets in $\mathbb{P}^n$) We say a subset $Y \subseteq \mathbb{P}^n$ is an algebraic set if $Y = Z(T)$ for some $T \subseteq S$. For an algebraic set Y , we define its homogeneous coordinate ring to be $S(Y) = S/I(Y)$.

- Notice that $Z(T) = Z(\langle T \rangle) = Z(\sqrt{\langle T \rangle})$ by Construction 2.3; hence Y is an algebraic set precisely when $I(Y) \subseteq S$ is a homogeneous ideal.

Proposition 2.4.1 (Zariski topology on $\mathbb{P}^n$) The finite union and arbitrary intersection of algebraic sets in $\mathbb{P}^n$ are again algebraic sets. We define the Zariski on $\mathbb{P}^n$ by declaring the closed sets to be precisely the algebraic sets.

- This is analogous to Proposition 1.2.

Definition 2.5 (Projective varieties) We say an algebraic set $Y \subseteq \mathbb{P}^n$ is a projective variety if Y is irreducible, and every open subset of Y is called a quasi-projective variety.

Construction 2.6 (Hyperplane) If $f \in S$ is a linear homogeneous polynomial, then $Z(f) \subseteq \mathbb{P}^n$ is called a hyperplane. In particular, we denote $Z(x_i)$ by H_i for $0 \leq i \leq n$. Notice that $U_i := \mathbb{P}^n \setminus H_i$ is an open set and $\mathbb{P}^n = \bigcup_{i=0}^n U_i$. For every i , we define

$$\varphi_i : U_i \rightarrow \mathbb{A}^n, \quad (a_0, \dots, a_n) \mapsto \left(\frac{a_0}{a_i}, \dots, \frac{a_n}{a_i} \right)$$

with a_i/a_i omitted. This is a well-defined function since $a_i \neq 0$ and the ratio a_j/a_i only depends on the equivalent class of $(a_1, \dots, a_n)$.

Proposition 2.6.1 (φ_i is a homeomorphism) *The map φ_i defined above is a homeomorphism of U_i with its induced topology to $\mathbb{A}^n$ with its Zariski topology.*

- WLOG we may assume $i = 0$ and write φ for φ_0 and U for U_0 . It is clear that φ is injective and admits an inverse, so it is bijective. To show bicontinuity, let S^h denote the set of homogeneous elements in S and consider

$$\alpha : S^h \rightarrow A := k[x_1, \dots, x_n], \quad f(x_0, \dots, x_n) \mapsto f(1, x_1, \dots, x_n).$$

Note that if $V := Z(f_1, \dots, f_m) \cap U$ is a closed set in U and $(a_0, \dots, a_n) \in V$, we may assume WLOG that $a_0 = 1$ up to rescaling since a_0 is non-zero. So then $\varphi(V) = Z(\alpha(f_1), \dots, \alpha(f_m)) \subseteq \mathbb{A}^n$ is an algebraic set and φ is closed. Conversely, consider

$$\beta : A \rightarrow S^h, \quad g(x_1, \dots, x_n) \mapsto x_0^{\deg g} g\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

Then if $Z(g_1, \dots, g_l) \subseteq \mathbb{A}^n$ is closed, so is $W := Z(\beta(g_1), \dots, \beta(g_l)) \subseteq \mathbb{P}^n$. Therefore $W \cap U$ is closed in U such that $\varphi^{-1}(Z(g_1, \dots, g_l)) = W \cap U$, since $\alpha \circ \beta = \text{id}_A$.

Corollary 2.6.2 (Affine components of a projective variety) *If $Y \subseteq \mathbb{P}^n$ is a projective (resp. quasi-projective) variety, then $Y = \bigcup_{i=0}^n Y \cap U_i$ where each $Y \cap U_i \subseteq \mathbb{A}^n$ is an affine (resp. quasi-affine) variety.*

- Indeed, notice that $Y \cap U_i$ is irreducible in U_i by Lemma 1.3.1, so it corresponds to an affine variety.

Exercise I.2

2.1 Prove the “homogeneous Nullstellensatz”: if $\mathfrak{a} \neq \bigoplus_{d>0} S_d$ is a homogeneous ideal, then $I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.

Proof. This follows immediately from the standard Nullstellensatz and the fact that the radical of a homogeneous ideal is again homogeneous. More concretely, using Corollary 2.6.2 we have

$$I(Z(\mathfrak{a})) = I\left(\bigcup_{i=0}^n Z(\mathfrak{a}_i)\right) \stackrel{(*)}{=} I(Z(\mathfrak{a}_0 \cdots \mathfrak{a}_n)) = \sqrt{\mathfrak{a}_0 \cdots \mathfrak{a}_n} = \sqrt{\mathfrak{a}}$$

where in $(*)$ we consider the extension $\langle \mathfrak{a}_i \rangle \trianglelefteq S$ for every i . □

2.2 Let $\mathfrak{a} \trianglelefteq S$ be a homogeneous ideal. Show that the TFAE:

- (i) $Z(\mathfrak{a}) = \emptyset$;
- (ii) $\sqrt{\mathfrak{a}} = S$ or $\sqrt{\mathfrak{a}} = S_+ := \bigoplus_{d>0} S_d$;
- (iii) $\mathfrak{a} \supseteq S_d$ for some $d > 0$.

Proof. (i) $\implies$ (ii) This is clear.

(ii) $\implies$ (iii) By Exercise 2.1, $\mathfrak{a} \supseteq S_d \iff \sqrt{\mathfrak{a}} \supseteq I(Z(S_d)) = \sqrt{\langle S_d \rangle}$; but $\sqrt{\langle S_d \rangle} \subseteq S_+ \subseteq \sqrt{\mathfrak{a}}$ if $d > 0$.

(iii) $\implies$ (i) Since $d > 0$ and $S = k[x_0, \dots, x_n]$ is an integral domain, if $\mathfrak{a} \supseteq S_d$ then $\sqrt{\mathfrak{a}} \supseteq S_1 \supseteq \{x_0, \dots, x_n\}$. Hence the zero locus of $\mathfrak{a}$ in $\mathbb{A}^{n+1}$ is just $\{0\}$ which is not a coset in $\mathbb{P}^n$, so we are done. $\square$

- In fact, for any graded domain $S = \bigoplus_{d \in \mathbb{N}} S_d$, we have that $\sqrt{S_d} = \bigcup_{e|d} S_e$.

2.3 (a) If $T_1 \subseteq T_2$ are subsets of S^h , then $Z(T_2) \subseteq Z(T_1)$.

(b) If $Y_1 \subseteq Y_2$ are subsets of $\mathbb{P}^n$, then $I(Y_2) \subseteq I(Y_1)$.

(c) For any subsets $Y_1, Y_2 \subseteq \mathbb{P}^n$, $I(Y_1) \cap I(Y_2) = I(Y_1 \cup Y_2)$.

(d) If $\mathfrak{a} \triangleleft S$ is a homogeneous polynomial such that $Z(\mathfrak{a}) \neq \emptyset$, then $I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.

(e) For any subset $Y \subseteq \mathbb{P}^n$, $Z(I(Y)) = \overline{Y}$.

Proof. (a) - (d) are clear. For (e), notice that $Z(I(Y)) \supseteq \overline{Y}$ since $Z(I(Y))$ is a closed set containing Y . Conversely, if $p \notin \overline{Y}$, then for some closed set $Y' \supseteq Y$, $p \notin Y'$. Hence $p \notin Z(I(Y')) \supseteq Z(I(Y))$. $\square$

2.4 (a) Let $\text{Rad}^h S$ denote the set of all homogeneous radical ideal in S . Then there is a bijection

$$\text{Rad}^h S \setminus \{S_+\} \leftrightarrow \{\text{algebraic sets in } \mathbb{P}^n\}.$$

(b) There is a bijection

$$\text{Spec}^h S \setminus \{S_+\} \leftrightarrow \{\text{projective varieties in } \mathbb{P}^n\}.$$

(c) $\mathbb{P}^n$ is irreducible.

Proof. (a) This is clear from Exercise 2.3(d)(e).

(b) Recall that the product and radical of homogeneous ideals are again homogeneous ideals, so this is completely analogous to the proof of Corollary 1.4.1.

(c) Note that $0 \triangleleft S$ is a homogeneous prime ideal. $\square$

2.5 (a) $\mathbb{P}^n$ is a Noetherian space.

(b) Every algebraic set in $\mathbb{P}^n$ can be written uniquely as a finite union of projective varieties, none containing another, which are called its irreducible components.

Proof. (a) This is clear since every descending chain of algebraic sets in $\mathbb{P}^n$ corresponds to an ascending chain of (homogeneous) ideals in $S = k[x_0, \dots, x_n]$ which is a Noetherian ring.

(b) This is immediate from Proposition 1.6.2. $\square$

2.6 Let $Y \subseteq \mathbb{P}^n$ be a projective variety. Show that $\dim S(Y) = \dim Y + 1$.

Proof. For every i , define $Y_i := \varphi_i(Y \cap U_i) \subseteq \mathbb{A}^n$. We claim that $A(Y_i) \simeq (S(Y)_{x_i})_0$. For clarity, assume $i = 0$. We start by defining a map $\alpha : (S_{x_0})_0 \rightarrow A(Y_0)$. Since every element in $(S(Y)_{x_0})_0$ is a finite sum of monomials of the form $f/x_0^{\deg f}$, we let $\alpha(f/x_0^{\deg f}) := \bar{f}(1, x_1, \dots, x_n) \in A(Y_0)$ and extend α linearly to $(S_{x_0})_0$. Notice that if $f \in I(Y)$, then $\alpha(f) \in I(Y_0)$. Hence we obtain an injective map $(S(Y)_{x_0})_0 \rightarrow A(Y_0)$ which we continue to denote by α . Now we show that it admits an inverse. Indeed, if $g \in A(Y_0)$ is a monomial, we define $\beta : A(Y_0) \rightarrow (S(Y)_{x_0})_0$ by $\beta(g) := g(x_1/x_0, \dots, x_n/x_0)$ and extend it to $A(Y_0)$ similarly. It is clear that $\beta \circ \alpha$ is the identity map, so the claim is proven and $A(Y_i) \simeq (S(Y)_{x_i})_0$ for every i . In particular,

$$A(Y_i)[x_i, x_i^{-1}] \simeq (S(Y)_{x_i})_0[x_i, x_i^{-1}] = S(Y)_{x_i}.$$

But $A(Y_i)[x_i, x_i^{-1}]$ is a finitely generated k -domain if $Y_i \neq \emptyset$, so

$$\dim S(Y)_{x_i} = \dim A(Y_i)[x_i, x_i^{-1}] = \text{trdeg}_k A(Y_i)[x_i, x_i^{-1}] = \text{trdeg}_k A(Y_i) + 1 = \dim A(Y_i) + 1.$$

Finally, we claim that $\dim S(Y)_{x_i} = \dim S(Y)$ for every i . To this end, it suffices to show that $x_i \notin J(S(Y)) := \bigcap_{\mathfrak{m} \in \text{Max } S(Y)} \mathfrak{m}$. But by a result from commutative algebra, $x_i \in J(S(Y)) \iff \forall f \in S(Y), 1 - x_i y \in S(Y)^\times$. Hence there exists a maximal ideal $\mathfrak{m} \not\ni x_i$. Since $S(Y)$ is a finitely generated k -domain, we deduce

$$\dim S(Y) = \text{ht}(\mathfrak{m}) = \dim S(Y)_{x_i} = \dim A(Y_i) + 1 = \dim Y_i + 1.$$

In particular, $\dim Y_i = \dim Y$ for every $Y_i \neq \emptyset$, and $\dim S(Y) = \dim Y + 1$. $\square$

2.7 (a) $\dim \mathbb{P}^n = n$.

(b) If $Y \subseteq \bar{Y}$ is a quasi-projective variety, then $\dim Y = \dim \bar{Y}$.

Proof. (a) This is immediate from Exercise 2.6 since $\dim \mathbb{P}^n + 1 = \dim S(\mathbb{P}^n) = n + 1$.

(b) Notice that $\dim Y = \sup \dim_i Y_i$ where $Y_i = \varphi_i(Y \cap U_i)$. But each $Y_i \subseteq \bar{Y}_i$ is a quasi-affine variety and therefore $\dim Y_i = \dim \bar{Y}_i$, so we are done. $\square$

2.8 A projective variety $Y \subseteq \mathbb{P}^n$ has dimension $n - 1 \iff Y = Z(f)$ for some irreducible homogeneous polynomial. In this case, we say Y is a hypersurface in $\mathbb{P}^n$.

Proof. This is clear from Exercise 2.6 and Proposition 1.7.3, since each non-empty $Y_i = Z(\alpha_i(f))$ is a hypersurface in $\mathbb{A}^n$ where α_i is analogous to α defined in the proof of Exercise 2.6. $\square$

2.9 *Projective closure of an affine variety.* If $Y \subseteq \mathbb{A}^n \simeq U_0$ is an affine variety, we denote by $\bar{Y}$ the closure of Y in $\mathbb{P}^n$ and call it the *projective closure* of Y .

(a) Show that $I(\bar{Y}) = \langle \beta(I(Y)) \rangle$ where β is as in the proof of Proposition 2.6.1.

(b) Let Y be the twisted cubic curve in Exercise 1.2. Find the generators for $I(Y)$ and $I(\bar{Y})$; use this example to show that for an affine variety Y , if $f_1, \dots, f_r$ generate $I(Y)$, then $\beta(f_1), \dots, \beta(f_r)$ do *not* necessarily generate $I(\bar{Y})$.

Proof. (a) Recall that $\beta : A \rightarrow S^h$ is given by $\beta(f) = x_0^{\deg f} f(x_1/x_0, \dots, x_n/x_0)$. Since

$$I(\bar{Y}) = \langle \beta(I(Y)) \rangle \iff \bar{Y} = Z(\langle \beta(I(Y)) \rangle),$$

it suffices to show that $Z(\langle \beta(I(Y)) \rangle)$ is the smallest closed set containing Y in $\mathbb{P}^n$. Now an element in $\langle \beta(I(Y)) \rangle$ is of the form $\sum_{i=1}^m g_i \beta(f_i)$ where $g_i \in S$ and $f_i \in I(Y)$. Notice that if $p \in \varphi_0(Y) \subseteq U_0$, we may assume WLOG $p = (1, a_1, \dots, a_n)$, so it is clear that

$$\sum_{i=1}^m g_i(p) \beta(f_i)(p) = \sum_i g_i(p) f_i(a_1, \dots, a_n) = \sum_i g_i(p) f_i(\varphi_0^{-1}(p)) = 0$$

and hence $Y \subseteq Z$. On the other hand, we need to show if $Y' \subseteq \mathbb{P}^n$ is a closed set containing Y , then $Y' \supseteq Z$. But

$$Y' \supseteq Y \iff I(Y') \subseteq \langle \beta(I(Y)) \rangle \implies Y' \supseteq Z(\langle \beta(I(Y)) \rangle) = Z.$$

(b) Recall that $I(Y) = \langle x^2 - y, x^3 - z \rangle \in \text{Spec } k[x, y, z]$. We claim that $\beta(xy - z) \notin \langle \beta(x^2 - y, x^3 - z) \rangle =: \mathfrak{b}$. Indeed, suppose to the contrary $\beta(xy - z) = xy - zt \in \mathfrak{b}$, then “evaluating” at $t = 0$ we deduce

$$xy - zt \equiv xy \in \bar{\mathfrak{b}} = \overline{\langle x^2 - yt, x^3 - z^2 \rangle} \equiv \langle x^2, x^3 \rangle \pmod{\langle t \rangle}$$

which is a contradiction since $xy \notin \langle x^2, x^3 \rangle$. Hence for $f_1, \dots, f_r \in A$, $\langle \beta(f_1), \dots, \beta(f_r) \rangle$ need not be the same as $\langle \beta(\langle f_1, \dots, f_r \rangle) \rangle$. $\square$

- To identify the specific generators for $\langle \beta(\langle x^2 - y, x^3 - z \rangle) \rangle$ will require a technique called Gröbner’s basis which we skip here. The fundamental reason behind the failure in (b) is that β is *not* a ring homomorphism (since $\beta(f + g) \neq \beta(f) + \beta(g)$). On the other hand, it is *true* that for a ring homomorphism $\varphi : A \rightarrow B$, $\langle a_1, \dots, a_n \rangle^e = \langle \varphi(a_1), \dots, \varphi(a_n) \rangle$.

2.10 *Cone over a projective variety.* Let $Y \subseteq \mathbb{P}^n$ be a non-empty algebraic set and $\theta : \mathbb{A}^{n+1} \setminus \{0\} \rightarrow \mathbb{P}^n$ be the map which sends a point to its homogeneous equivalent class. We define the *affine cone* over Y to be $C(Y) := \theta^{-1}(Y) \cup \{0\}$.

- Show that $C(Y) \subseteq \mathbb{A}^{n+1}$ is an algebraic set with $I(C(Y)) = I(Y)$, where $I(Y)$ is considered as an ordinary variety in $k[x_0, \dots, x_n]$.
- $C(Y)$ is irreducible $\iff Y$ is.
- $\dim C(Y) = \dim Y + 1$.

Proof. (a) Write $Y = Z(f_1, \dots, f_m)$ for some $f_i \in S^h$. Since $Y \neq \emptyset$, we must have $\deg f_i > 0$ for every i ; otherwise $Z(f_i) = \emptyset$. We claim that $C(Y) = Z(f'_1, \dots, f'_m)$ where $f'_i = f_i$ but we view them as functions over $\mathbb{A}^{n+1}$. Since $\deg f_i > 0$ and f_i is homogeneous, $f_i(0) = 0$. On the other hand, if $p \in \mathbb{A}^{n+1} \setminus \{0\}$, then

$$p \in C(Y) \iff p \in \theta^{-1}(Y) \iff \theta(p) \in Y \iff (\forall i) : f_i(\theta(p)) = 0 \xLeftrightarrow{(*)} (\forall i) : f_i(p) = 0$$

where $(*)$ holds because for f_i is homogeneous. So the claim is proven and $C(Y) = Z(f'_1, \dots, f'_m) \subseteq \mathbb{A}^{n+1}$. As a result, $C(Y)$ is an algebraic set with $I(C(Y)) = I(Y)$.

- This is immediate from (a) since $I(C(Y)) = I(Y)$.
- By Exercise 2.6, we have that

$$\dim C(Y) = \dim A(C(Y)) = \dim k[x_0, \dots, x_n]/I(Y) = \dim S(Y) = \dim Y + 1.$$

$\square$

- Sometimes we also consider the projective closure $\overline{C(Y)}$ of $C(Y)$ over $\mathbb{P}^{n+1}$. This is called the *projective cone* over Y .

2.11 *Linear varieties in $\mathbb{P}^n$.* A hypersurface defined by a linear polynomial is called a hyperplane.

- If $Y \subseteq \mathbb{P}^n$ is a variety, then Y is an intersection of hyperplanes $\iff I(Y)$ is generated by linear polynomials.
- If $Y \subseteq \mathbb{P}^n$ is a linear variety of dimension r , then $I(Y)$ is minimally generated by $n - r$ elements.
- Let $Y, Z \subseteq \mathbb{P}^n$ be linear varieties, with $\dim Y = r$ and $\dim Z = s$. If $r + s \geq n$, then $Y \cap Z \neq \emptyset$. Furthermore, if $Y \cap Z \neq \emptyset$, then $Y \cap Z$ is a linear variety of dimension at least $r + s - n$.

Proof. (a) If f_i are linear polynomials, then

$$Y = \bigcap_{i \in I} Z(f_i) = Z(f_i : i \in I) \iff I(Y) = I(Z(f_i : i \in I)) = \sqrt{\langle f_i : i \in I \rangle} =: \sqrt{\mathfrak{a}}.$$

We claim that $\mathfrak{a} \triangleleft S := k[x_0, \dots, x_n]$ is prime. Indeed, consider the subspace $V := \text{span}_k\{f_i : i \in I\} \subseteq S_1$ where S_1 is a $(n+1)$ -dimensional k -vector space. Then we can find a k -basis $\{f_0, \dots, f_m\}$ for V and therefore $\mathfrak{a} = \langle f_0, \dots, f_j \rangle$. Now since the f_j 's are linearly independent, up an automorphism of $k[x_0, \dots, x_n]$ we may assume $f_j = x_j$ for every $j \leq m$. Hence

$$S/\mathfrak{a} = k[x_0, \dots, x_n]/\langle x_0, \dots, x_m \rangle \simeq k[x_{m+1}, \dots, x_n]$$

is an integral domain, so the claim is proven. In particular, an algebraic set Y is a linear variety $\iff Y$ is an intersection of hyperplanes.

(b) Since $Y \subseteq \mathbb{P}^n$ is a variety of dimension r , $\dim S(Y) = r+1$. Hence

$$\text{ht}(I(Y)) + \dim S(Y) = \dim S = n+1 \implies \text{ht}(I(Y)) = n-r,$$

so by Krull's HT, $n-r = \text{ht}(I(Y)) \leq d(I(Y))$. Now we show $I(Y)$ can be generated by $n-r$ elements if Y is a linear variety. Indeed, by (a) we can write $I(Y) = \langle f_0, \dots, f_m \rangle$ for some minimal $m \in \mathbb{N}^{>0}$ and linear polynomials f_i . Hence up to some $\varphi \in \text{Aut}_k(S)$, we have that

$$\begin{aligned} r+1 = \dim Y + 1 = \dim S(Y) &= \dim \varphi(k[x_0, \dots, x_n])/I(Y) \\ &= \dim k[f_0, \dots, f_m, x_{m+1}, \dots, x_n]/\langle f_0, \dots, f_m \rangle = n-m. \end{aligned}$$

Therefore $m = n-r-1$ and $I(Y) = \langle f_0, \dots, f_m \rangle$ can be generated by $n-r$ elements, so we are done.

(c) Notice that $I(Y \cap Z) = \sqrt{I(Y) + I(Z)} =: \sqrt{\mathfrak{a}}$. Since Y and Z are linear varieties, by (a) and (b) we deduce that $\mathfrak{a}$ can be generated by linear polynomials where

$$d(\mathfrak{a}) \leq d(I(Y)) + d(I(Z)) = (n-r) + (n-s) \leq n < n+1 = \dim_k S_1.$$

Let $\mathfrak{a}_1$ denote the subset of all linear polynomials in $\mathfrak{a}$, then $\mathfrak{a}_1 \subsetneq S_1$ is a proper subspace. In particular, we must have that $\sqrt{\mathfrak{a}_1} \not\supseteq S_1$ since $\sqrt{\mathfrak{a}_1} = \mathfrak{a}_1$. As a result, $I(Y \cap Z) = \sqrt{\mathfrak{a}} \neq S$ and therefore $Y \cap Z \neq \emptyset$. Furthermore, if $Y' := Y \cap Z \neq \emptyset$, then Y' is a linear variety by (a), where $d(I(Y')) \leq 2n - (r+s)$. Hence

$$\dim Y' = \dim S(Y') - 1 = (n+1) - \text{ht}(I(Y')) - 1 = n - \text{ht}(I(Y')) \geq r+s-n.$$

□

2.12 *d-uple embedding.* For $n, d > 0$, let $M_0, \dots, M_N$ be all the monomials of degree d in $n+1$ variables $x_0, \dots, x_n$, where $N = \binom{n+d}{d} - 1$. Define a map $\rho_d : \mathbb{P}^n \rightarrow \mathbb{P}^N$ by $\rho_d(p) := (M_0(p), \dots, M_N(p))$. This is called the *d-uple embedding* of $\mathbb{P}^n$ in $\mathbb{P}^N$; for instance, if $n = 1$ and $d = 2$, then $N = 2$ and the image of $\mathbb{P}^1$ in $\mathbb{P}^2$ is a conic.

(a) Define $\theta : k[y_0, \dots, y_N] \rightarrow k[x_0, \dots, x_n]$ by $\theta(y_i) := M_i$. Then $\mathfrak{a} := \ker \theta$ is a homogeneous prime ideal and so $Z(\mathfrak{a}) \subseteq \mathbb{P}^N$ is a projective variety.

(b) Show that $\text{im } \rho_d = Z(\mathfrak{a})$.

(c) Show that ρ_d is a homeomorphism from $\mathbb{P}^n$ to $Z(\mathfrak{a})$.

(d) Show that the twisted cubic curve in $\mathbb{P}^3$ is equal to the 3-uple embedding of $\mathbb{P}^1$ in $\mathbb{P}^3$.

Proof. (a) First we show that $\mathfrak{a}$ is a homogeneous ideal. To this end, it suffices to show that if $f \in \mathfrak{a}$, then all homogeneous components of f are in $\mathfrak{a}$. Suppose $f = \sum_i f_i$ where f_i is of homogeneous degree i ; then $\theta(f_i)$ is of homogeneous degree id . As a result,

$$0 = \theta(f) = \sum_i \theta(f_i) \in \bigoplus_i S_{id} \implies (\forall i) : \theta(f_i) = 0.$$

Hence $\mathfrak{a}$ is a homogeneous ideal. Furthermore, if $fg \in \mathfrak{a}$, then $\theta(f)\theta(g) = \theta(fg) = 0$, which implies that $\theta(f) = 0$ or $\theta(g) = 0$ since $k[x_0, \dots, x_n]$ is an integral domain. So $\mathfrak{a}$ is also prime and therefore $Z(\mathfrak{a}) \subseteq \mathbb{P}^N$ is a projective variety.

(b) That $\text{im } \rho_d \subseteq Z(\mathfrak{a})$ is clear: if $f \in \mathfrak{a}$ and $p \in \mathbb{P}^n$, then

$$f(\rho_d(p)) = f(M_0(p), \dots, M_N(p)) = f(M_0, \dots, M_N)(p) = \theta(f)(p) = 0.$$

For the other inclusion, suppose $q \in Z(\mathfrak{a})$; we construct a preimage of q . To this end, for every $M \in \{M_i\}$ let $y_M \in \{y_i\}$ denote the variable such that $\theta(y_M) = M$. We claim that $y_{x_i^d}(q) \neq 0$ for some i . Indeed, notice that for every M ,

$$M^d = \prod_{i=0}^n (x_i^d)^{\deg_{x_i}(M)} \implies (y_M)^d \equiv \prod_{i=0}^n (y_{x_i^d})^{\deg_{x_i}(M)} \pmod{\mathfrak{a}}.$$

where $\deg_{x_i}(M)$ denotes the power of x_i in M . Since $q \in Z(\mathfrak{a})$, if $y_{x_i^d}(q) = 0$ for every i , then $y_M(q)^d = 0$ for every M , contradicting that $q \in \mathbb{P}^N$. As a result, we may assume WLOG that $y_{x_0^d}(q) = 1$. Now for every M , we have that

$$(y_{x_0^d})^{d-1} y_M \equiv \prod_{i=0}^n (y_{x_i^{d-1}x_0})^{\deg_{x_i}(M)} \pmod{\mathfrak{a}},$$

so then

$$y_M(q) = 1^{d-1} \cdot y_M(q) = \prod_{i=0}^n y_{x_i^{d-1}x_0}(q)^{\deg_{x_i}(M)}.$$

Hence we can “parametrize” every coordinate of q using $\{y_{x_i^{d-1}x_0}(q)\}$; in other words, to construct a preimage for $q \in Z(\mathfrak{a})$, it suffices to find $p \in \mathbb{P}^n$ such that $y_{x_i^{d-1}x_0}(\rho_d(p)) = y_{x_i^{d-1}x_0}(q)$ for every i . But then we can simply consider

$$p := (y_{x_0^d}(q), y_{x_1^{d-1}x_0}(q), \dots, y_{x_n^{d-1}x_0}(q)) \in \mathbb{P}^n$$

which is well-defined since $y_{x_0^d}(q) \neq 0$, so we are done.

(c) (INCOMPLETE) It remains to show that ρ_d is bicontinuous. Suppose $Y \subseteq \mathbb{P}^n$ is a projective variety. We claim that $\rho_d(Y) = Z(\theta^{-1}(I(Y)))$. Indeed, by the same reasoning as in (b) it is clear that $\rho_d(Y) \subseteq Z(\theta^{-1}(I(Y)))$. (Is the argument completely analogous to (b) with Y replacing $\text{im } \rho_d$ and $Z(\theta^{-1}(I(Y)))$ replacing $Z(\mathfrak{a})$?)

(d) (INCOMPLETE) If $n = 1$ and $d = 3$, then $\rho_d(p) = (x_1^3, x_1^2x_2, x_1x_2^2, x_2^3)$. Let $Y \subseteq \mathbb{P}^3$ be the twisted cubic curve and recall that

$$I(Y) = \langle x^2 - yt, x^3 - zt^2, xy - z \rangle \triangleleft k[t, x, y, z].$$

By (a), we have that $\rho_d(\mathbb{P}^1) = Z(\mathfrak{a})$; so to show that $\rho_d(\mathbb{P}^1) = Y$, it suffices to show $\mathfrak{a} = I(\rho_d(\mathbb{P}^1)) = I(Y)$. That $I(Y) \subseteq \mathfrak{a}$ is clear since

$$\theta(I(Y)) = \langle (x_0^2x_1)^2 - x_0x_1^2 \cdot x_0^3, (x_1^2x_1)^3 - x_1^3 \cdot x_0^6, x_0^2x_1 \cdot x_0x_1^2 - x_0^3x_1^3 \rangle = 0.$$

On the other hand, if $f \in \mathfrak{a}$, then ... □

- The moral of this question is that if the ideal of a projective variety Y is given by the kernel of a k -algebra homomorphism θ , then Y is topologically associated with the underlying space of θ 's domain. See Exercise 2.14 for another example.

2.13 Let Y be the image of the 2-uple embedding of $\mathbb{P}^2$ in $\mathbb{P}^5$, which is called the *Veronese surface*. If $Z \subseteq Y$ is a curve, show that there exists a hypersurface $V \subseteq \mathbb{P}^5$ such that $V \cap Y = Z$.

2.14 *Segre embedding.* Let $\psi : \mathbb{P}^r \times \mathbb{P}^s \rightarrow \mathbb{P}^N$ be the map sending the ordered pair $(a_0, \dots, a_r) \times (b_0, \dots, b_s)$ to the point $(\dots, a_i b_j, \dots)$ in lexicographic order, where $N = rs + r + s$. Then ψ is well-defined and injective; it is called the *Segre embedding*. Show that $\text{im } \psi \subseteq \mathbb{P}^n$ is a projective variety.

Proof. Suppose the homogeneous coordinate of $\mathbb{P}^N$ is given by $\{z_{ij} : (i, j) \in [0, r] \times [0, s]\}$ and let $\mathfrak{a}$ be the kernel of the homomorphism $\theta : k[\{z_{ij}\}] \rightarrow k[x_0, \dots, x_r, y_0, \dots, y_s]$ where $\theta(z_{ij}) = x_i y_j$. We claim that $\text{im } \psi = Z(\mathfrak{a})$. One inclusion is easy: if $(a, b) \in \mathbb{P}^r \times \mathbb{P}^s$ and $f \in Z(\mathfrak{a})$, then

$$f(\psi(a, b)) = f(\dots, a_i b_j, \dots) = f(\dots, x_i y_j, \dots)(a, b) = \theta(f)(a, b) = 0,$$

so $\text{im } \psi \subseteq Z(\mathfrak{a})$. For the other inclusion, suppose $p \in Z(\mathfrak{a})$; we construct a preimage for p . To this end, we may assume WLOG that $z_{00}(p) = 1$. Then since $p \in Z(\mathfrak{a})$, for all $i, j > 0$ we deduce

$$z_{00} z_{ij} \equiv z_{i0} z_{0j} \pmod{\mathfrak{a}} \implies z_{ij}(p) = 1 \cdot z_{i0}(p) z_{0j}(p).$$

Hence we can “parametrize” the coordinate of p using $\{z_{i0}(p), z_{0j}(p) : (i, j) \in [1, r] \times [1, s]\}$; in other words, to find the preimage of p , it suffices to construct $(a, b) \in \mathbb{P}^r \times \mathbb{P}^s$ such that for all $i, j \geq 0$,

$$a_i b_0 = z_{i0}(\psi(a, b)) = z_{i0}(p) \quad \text{and} \quad a_0 b_j = z_{0j}(\psi(a, b)) = z_{0j}(p).$$

Notice that we may assume WLOG that $a_0 = b_0 = 1$ since $a_0 b_0 = z_{00}(p) = 1$. As a result, we may simply let $a_i := z_{i0}(p)$ and $b_j := z_{0j}(p)$. It is clear that $(a, b) \in \mathbb{P}^r \times \mathbb{P}^s$ is well-defined, so we are done. $\square$

2.15 *Quadratic surfaces in $\mathbb{P}^3$.* Consider the surface $Q \subseteq \mathbb{P}^3$ defined by the equation $xy - zw = 0$.

- (a) Show that Q is equal the Segre embedding of $\mathbb{P}^1 \times \mathbb{P}^1$ in $\mathbb{P}^3$, for suitable choice of coordinates.
- (b) Show that Q contains two families of lines $\{L_t\}, \{M_t\}$, each parametrized by $t \in \mathbb{P}^1$, with the properties that if $L_t \neq L_u$, then $L_t \cap L_u = \emptyset$; if $M_t \neq M_u$, then $M_t \cap M_u = \emptyset$; and for all t, u , $L_t \cap M_u$ is a singleton.
- (c) Show that Q contains other curves besides these lines, and deduce that the Zariski topology on Q is not homeomorphic to the product topology on $\mathbb{P}^1 \times \mathbb{P}^1$ via the Segre embedding.

- 2.16** (a) The intersection of varieties need not be a variety. For example, suppose $Q_1, Q_2 \subseteq \mathbb{P}^3$ are given by the equations $x^2 - yw = 0$ and $xy - zw = 0$, respectively. Show that $Q_1 \cap Q_2$ is the union of a twisted cubic curve and a line.
- (b) Even if the intersection of two varieties is a variety, the ideal of the intersection may not be sum of the ideals. For example, let C be the conic in $\mathbb{P}^2$ given by the equation $x^2 - yz = 0$. Let L be the line given by $y = 0$. Show that $C \cap L$ consists of a point p , but $I(C) \cap I(L) \neq I(p)$.

2.17 *Complete intersections.* A variety Y of dimension r in $\mathbb{P}^n$ is a (strict) *complete intersection* if $I(Y)$ can be generated by $n - r$ elements. Y is a *set-theoretic complete intersection* if Y can be written as the intersection of $n - r$ hypersurfaces.

- (a) Let $Y = Z(\mathfrak{a}) \subseteq \mathbb{P}^n$ be a variety. Suppose $d(\mathfrak{a}) \leq q$. Show that $\dim Y \geq n - q$.
- (b) Show that a strict complete intersection is a set-theoretic complete intersection.
- (c) The converse of (b) is false. For example let $Y \subseteq \mathbb{P}^3$ be the twisted cubic curve. Show that $I(Y)$ cannot be generated by two elements; on the other hand, find hypersurfaces H_1, H_2 of degrees 2, 3 respectively, such that $Y = H_1 \cap H_2$.

Morphisms